# THE FUNCTOR ASSOCIATING A QUASI-COHERENT MODULE ON A SCHEME TO A MODULE ON ITS SITE IS FULLY FAITHFUL

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The statement refers to the functors of a same-chapter definition; that definition environment is delivered as a context span so that the reader can see the functor being spoken about, and it is not counted as a statement.

*Editorial note.* The proof opens with a parenthetical citation of the lemma showing the functors exist; the parenthetical is reproduced verbatim.

*Editorial note.* The counted spans carry 4 navigation stubs; no bibliography citation occurs in the spans.

*Editorial note.* The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

*Allowed standard prerequisites (editorial).*

- The definition of the associated module F^a on a site. Same chapter, Definition definition-structure-sheaf, invoked in the statement at lines 2680-2681 and delivered as a context span. Fixes the functor whose full faithfulness is asserted.
- The associated module of a quasi-coherent module is quasi-coherent on the site. Same chapter, Lemma lemma-quasi-coherent-gives-quasi-coherent, invoked at proof lines 2691-2692. Shows that the functors are indeed functors between the categories named in the statement.
- The association is pullback along a morphism of ringed sites and the comparison of topologies. Same chapter, Remark remark-change-topologies-ringed-sites at proof lines 2695-2697 and Topologies, Section topologies-section-change-topologies at proof lines 2700-2702. Supplies the morphism of ringed sites and the description of its pushforward as restriction along the inclusion.

*Source reference.* Original definition 03DU: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 949–949. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original lemma 03DV: sources/stacks-descent/pinned-original-source.tex, source lines 2616–2616. Pointer outside the counted span, invoked once

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for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original remark 03FH: sources/stacks-descent/pinned-original-source.tex, source lines 2587–2587. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original section 03FE: sources/stacks-chow/tags.txt, source lines 4489–4489. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

**Lemma 0.1.** *Let $S$ be a scheme. Let $\tau \in \{Zariski, étale, smooth, syntomic, fppf\}$. Each of the functors $\mathcal{F} \mapsto \mathcal{F}^a$ of Definition 03DU*

$$QCoh(\mathcal{O}_S) \to QCoh((Sch/S)_\tau, \mathcal{O}) \quad or \quad QCoh(\mathcal{O}_S) \to QCoh(S_\tau, \mathcal{O})$$

*is fully faithful.*

*Proof.* (By Lemma 03DV we do indeed get functors as indicated.) We may and do identify $\mathcal{O}_S$-modules on $S$ with modules on $(S_{Zar}, \mathcal{O}_S)$. The functor $\mathcal{F} \mapsto \mathcal{F}^a$ on quasi-coherent modules $\mathcal{F}$ is given by pullback by a morphism $f$ of ringed sites, see Remark 03FH. In each case the functor $f_*$ is given by restriction along the inclusion functor $S_{Zar} \to S_\tau$ or $S_{Zar} \to (Sch/S)_\tau$ (see discussion of how these morphisms of sites are defined in Topologies, Section 03FE). Combining this with the description of $f^*\mathcal{F} = \mathcal{F}^a$ we see that $f_*f^*\mathcal{F} = \mathcal{F}$ provided that $\mathcal{F}$ is quasi-coherent. Then we see that

$$\mathrm{Hom}_{\mathcal{O}}(\mathcal{F}^a, \mathcal{G}^a) = \mathrm{Hom}_{\mathcal{O}}(f^*\mathcal{F}, f^*\mathcal{G}) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{F}, f_*f^*\mathcal{G}) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{F}, \mathcal{G})$$

as desired. □